

SCIENCE IS THE PATH TO KNOWLEDGE, AND FINDING OUT ABOUT HOW OUR UNIVERSE WORKS. YOU CAN'T BE A GEEK AND NOT KNOW YOUR SCIENCE!

THIS MONTH IN SCIENCE:

In Earth, explore areas on our planet that are similar to outworldly terrain. In Cutting Edge, we talk about some 'Roach' tech and in Health, we discuss how AI is helping in the fight against COVID-19.



Spotty hot stars

Massive and energetic stars called extreme horizontal branch stars are plagued with giant magnetic spots as well as superflare events millions of times more powerful than similar explosions seen on the Sun. These are peculiarly hot objects that tend to die out prematurely. <https://digit.in/hotstars>

WHAT'S NEW

Tiny grains of sand move massive glaciers

During glacial periods, sheets of ice extend from the glaciers in the polar regions and high up in the mountains to cover vast regions of the planet. In previous ice ages, these sheets have spread across all of North America, Europe and Siberia. This month's *dmystify* has more details about these glacialations. The planet is currently in an interglacial period, between two ice ages, and glaciers cover about 10 percent of the surface of the planet. Normally, these glaciers creep across the Earth, but at times there can be sudden surges where they move across the surface at 10 to 100 times



their regular speeds. These glacial surges can cause avalanches, create floods in rivers, and overwhelm settlements in their paths.

Researchers have long been mystified about what exactly causes these glaciers to surge, and now they might have an answer. These glacial surges are believed to be caused by tiny grains of sand that are beneath the massive walls of ice that form the glaciers. <https://digit.in/fastglacier>

Understanding satellite health using sunlight

Those operating satellites from the ground can understand from the behavior of the satellite that it is experiencing issues, maybe the batteries are not getting charged properly from the solar panels or the orbit is unstable. However, it is very difficult to get an outside perspective of a deployed satellite, the same way someone would go out and check a vehicle after a scratch or a bump. A team of



researchers from Purdue are doing just that using ground based telescopes. Satellites are typically deployed at such an altitude that they nearly always catch the Sun and are

visible as tiny points of light, like fast moving stars. By analysing the characteristics of the light on the single pixel over time, scientists can diagnose problems that the satellite may be having. This observation is called the light curve of the object. By studying the light curves of satellites, scientists have developed an approach that can inform them about the health of the satellite.

<https://digit.in/junksat>



70 Trillion FPS

Researchers from Caltech have developed a camera that can capture images at 70 trillion frames per second, which is fast enough to capture waves of light in motion and the fluorescent decay of molecules. The camera can help in research on fundamental physics. <https://digit.in/caltechfps>



Graphene clothing

A team of scientists from Manchester University have developed a smart adaptive textile that can keep the body temperature of the wearer cool in hot conditions. The textile transitions between a heat retaining state and a heat losing state depending on the temperature. <https://digit.in/gphclth>



Turbulent lasers

Just like air and water, light can be turbulent too and researchers from Harvard have used this property to create a precision laser system that can be used to detect and measure different frequencies of light with precision. The research has applications in sensing and spectroscopy. <https://digit.in/lascomb>